# Combining Queries in EF Core

Where() + Select() + OrderBy()

## 1. Core idea

You can chain filtering, projection, and sorting into one LINQ query. EF Core translates the composed query into SQL, and the database performs those operations when the query executes.

## 2. What each operation does

| Operation | Purpose | Typical SQL |
|---|---|---|
| **Where()** | Filters rows that match a condition. | `WHERE` |
| **Select()** | Chooses which values/columns to return. | `SELECT` |
| **OrderBy()** / **OrderByDescending()** | Sorts the returned results. | `ORDER BY` |

**Mental model: Filter → choose columns → sort → execute. These LINQ operations are normally translated into one SQL query, not one query per method.**

## 3. Short example

Requirement: retrieve active students, return only their ID and full name, and show the newest IDs first.

FilteredStudents.cs

```
var query = context.Students
    .Where(s => s.Status == "Active")
    .Select(s => new
    {
        s.StudentId,
        FullName = s.FirstName + " " + s.LastName
    })
    .OrderByDescending(s => s.StudentId);

var students = query.ToList();
```

## 4. SQL shape

EF Core can translate the composed query to SQL similar to this:

GeneratedSQL.sql

```
SELECT [s].[StudentId],
       ([s].[FirstName] + N' ') + [s].[LastName] AS [FullName]
FROM [Students] AS [s]
WHERE [s].[Status] = N'Active'
ORDER BY [s].[StudentId] DESC;
```

The database filters rows, returns only selected columns, and sorts the result. `ToQueryString()` previews the SQL shape; runtime logging shows the command executed.

## 5. Avoid early execution

Calling `ToList()` before applying filtering or projection materializes the full result first. Later operations then run in memory and can transfer much more data than needed.

Prefer building the query first and calling `ToList()` at the point you actually need the results.

## 6. Practical pattern

**Where** → **Select** → **OrderBy** → **ToList** is a common pattern for list screens, dashboards, and API responses. The exact operator order can vary depending on the required result, but keep database-translatable operations before materialization.

Mental anchor: build the query completely first — then execute it.

Next lesson: checking existence with Any() and All().